

Question Paper

Exam Date & Time: 25-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Data Structures [ECM_2125]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Write a function Reverse to reverse the elements of an integer array using pointer. Access the elements of the array using dereference operator. Read the values from the keyboard in main function. Display the result in the main function. (4)
- A)

Course Outcome: "CO1" / Cognitive Level: "Application"

- B) Define integer pointer, character pointer and float pointer with examples. Write a short program to swap two integers using pass by reference method. (3)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- C) Determine and explain the functionality of the following code fragment and write the output of any sample input. (3)

```
void Funtion1(int a[], int n)
{
    for (int i = 0; i < n - 1; i += 2)
    {
        int temp = a[i];
        a[i] = a[i + 1];
        a[i + 1] = temp;
    }
}
```

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2) You are given the following sequence of letters and asterisks: (4)
- EAS*Y*QUE***ST***IO*N***

- A) Each **letter** (such as E) represents an **insert operation** of that element into a data structure, and each **asterisk (*)** represents a **remove operation**.

1. **Assume the data structure is a Stack (LIFO order)**. Show the sequence of elements that would be **removed** (output sequence) during the remove operations.
2. **Assume the data structure is a Queue (FIFO order)**. Show the sequence of elements that would be **removed** (output sequence) during the remove operations.
3. **Compare and explain** briefly how and why the two output sequences differ, even though the same sequence of operations is applied.

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- B) Write a C program that accepts integers continuously and inserts them into a stack until the user enters a negative number. Display all elements in the stack. (3)

Course Outcome: "CO2" / Cognitive Level: "Application"

- C) Convert the given infix expression to prefix expression. Show the content of Stack at every step: (3)
 $A + B * (C \wedge D - E) \wedge (F + G * H) - I$

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3) Write the structure definition for a singly linked list node in C and function definitions to perform the following operations: (4)

- A) i) Insert a new node at the end of the list
ii) Display the data of the middle node by traversing the list only once
iii) Display all node data in reverse order without actually reversing the list

Course Outcome: "CO3" / Cognitive Level: "Application"

- B) Write a C Program to do the following: (3)
i) Create a circular singly linked list where each node contains an integer value.
ii) Insert a new node with a specified value after every node containing an even integer.

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- C) write a c program to implement stack using linked list and implement push() and pop() operation with all necessary conditions (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4) Consider a following sequence 40, 20, 60, 10, 30, 50, 70, Construct a binary search tree (BST) by inserting the elements in the order given above. Draw the resulting BST. (4)

- A) 1. Write a C program to insert a new integer element into the BST while maintaining the BST property. The program should:

- Take the new element as input from the user
- Insert the element in the correct position in the BST
- Display the BST using inorder traversal after insertion

Write a recursive C function to search for a given integer element in the BST. The function should return whether the element is present or not.

Course Outcome: "CO4" / Cognitive Level: "Application"

- B) A circular queue of size 6 is initially empty. Perform the following operations: ENQUEUE(10), ENQUEUE(20), ENQUEUE(30), ENQUEUE(40), DEQUEUE(), ENQUEUE(50), ENQUEUE(60), ENQUEUE(70), DEQUEUE(), ENQUEUE(80) (3)

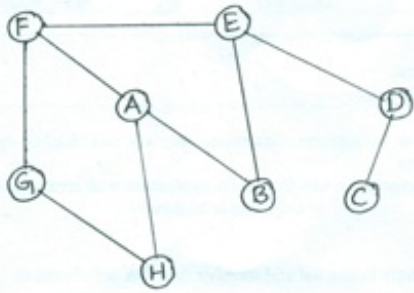
- i) Show the **contents of the queue** after each operation.
ii) Track **Front (F)** and **Rear (R)** pointers after every step.
iii) Identify the step where **overflow** occurs and justify your answer.

Course Outcome: "CO2" / Cognitive Level: "Understand"

- C) Given three sequences: (3)
(i) 28, 16, 11, 19, 17, 25
(ii) 11, 16, 17, 19, 25, 28
(iii) 11, 17, 25, 19, 16, 28
Determine which one is preorder, inorder, and postorder sequences of the same binary tree, and construct that tree.

- 5) Write a BFS Traversal sequence of the following Graph. Write the function to find the BFS taking as (4)
the parameter the value of the vertex from which BFS (depth first search) has to be carried out.

A)



- B) Define the following for the binary tree. (3)

1. What is n-ary tree.
2. Strict n-ary tree.
3. Height Vs. Node
4. Internal Vs. External Node

Indegree Vs Outdegree

- C) Construct a binary tree given its inorder and preorder sequence as B F G H P R S T W Y Z and P F (3)
B H G S R Y T W Z respectively. Also write the postorder traversal sequence for the constructed
binary tree.

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Event Name	Subject
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058-END SEMSTER-ODD-2025 ECM_2125:Data Structures

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	13
3	3	CO 3	10
4	4	CO 4	17
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 ECM_2125:Data Structures

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	4	12(24%)
3	Application	9	31(62%)
4	Analysis	2	7(14%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)